

# Impact of AMI Data Time Granularity on Quasi-Static Time-Series Load Flow Simulation

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**Abstract**—AMI data allows highly detailed representation of distribution system loads in quasi-static time-series load flow (QSTS) simulations but requires managing large data volumes. Reducing the time granularity of AMI data reduces the data volume but can harm the detail and accuracy of QSTS simulations. This paper analyzes the accuracy of QSTS simulations with different time granularities of AMI load profiles. Three time granularities (5-min, 15-min, and 60-min) common to today's AMI deployments are compared against a higher 1-min time granularity. The QSTS simulation results show that while lower time granularities of AMI data may capture feeder operation in general at the medium-voltage level, AMI data with high granularity is necessary to accurately capture feeder maximum element loadings and minimum node voltages particularly in the low-voltage secondary circuits.

**Index Terms**— Data Analytics, Load Modeling, Load Flow Analysis, Power Distribution, Smart Metering

## I. INTRODUCTION

Conventional distribution load modeling practices are characterized by limited visibility to loads downstream of substation[1-3]. Measurement data from AMI and other emerging distribution measurement sources provides a new level of visibility to distribution system loads. Among other applications, AMI data allows improved representation of the spatial and temporal diversity of distribution system loads in quasi-static time-series load flow (QSTS) simulations. However, leveraging AMI data for QSTS simulations requires handling large data volumes [4]. For example, 1-minute real and reactive power measurements for all customers on a single feeder for a year could be up to 10 GB.

The volume of AMI data required for QSTS simulations depends, among other things, on the length and time granularity of the QSTS simulations. The data requirements could be reduced by shortening the QSTS simulation time or by reducing the QSTS simulation time period. This however, can also reduce the accuracy and detail of the QSTS simulations. The impact of QSTS simulation time period and time granularity is analyzed in [5] but only using feederhead SCADA data as opposed to AMI data. The accuracy of QSTS simulations of two low-voltage secondary circuits with 15-minute and 1-minute time granularity is evaluated in [6]. This paper expands the previous research by comparing the accuracy of QSTS simulations with different time granularities of AMI load profiles. The accuracy of QSTS simulations with three time granularities (5-minute, 15-minute, and 60-minute) common to today's AMI deployments

are compared against a higher time granularity of 1-minute. The QSTS simulations' result dataset, which included over 4.5 billion data points of feeder node voltages and element phase currents, was analyzed in detail to statistically characterize the accuracy of QSTS simulations with different time granularities of AMI data.

The rest of this paper is organized as follows. Section II introduces the case study feeder model analyzed in this paper. Section III discusses some of the challenges experienced when performing QSTS simulations with high granularity AMI load profiles. Section IV discusses the simulation results and compares the results with 60-minute time granularity to the results with 1-minute time granularity. Section V compares the QSTS simulation accuracy with the four time granularities. Section VI concludes the paper.

## II. CASE STUDY FEEDER MODEL

### A. Distribution Feeder Model

In this paper, QSTS simulation accuracy is analyzed with different AMI load profile time granularities on a modified version of the OpenDSS model of EPRI "Ckt5" (Fig 1), which consists of a 5-km long 12.47 kV distribution feeder supplying 1,379 residential loads with 591 service transformers. The feeder model has no load tap changer or line voltage regulators but includes four switched-capacitor banks, which were disabled to focus on the impact of time granularity of AMI data, as opposed to the impact of mismatching controller states between simulations.

### B. Modeling Loads at 1-Minute Time Granularity

To analyze QSTS simulation accuracy with different time granularities of AMI load profiles, Ckt5 loads were first represented with high-granularity 1-minute residential customer load data obtained from Pecan Street Inc. [7]. As follows, the load modeling is discussed at a high level and for more details, the reader is referred to [8]. Moreover, as reactive power data was not available in the original Pecan Street dataset, reactive power loads were represented by assigning each load a random power factor between 0.79-0.99 that was kept constant for a duration of 30 minutes. This assumption was made to represent the diverse random behavior of reactive power of different feeder loads while still considering that load power factors tend to change relatively slowly. For the QSTS simulations, all loads were modeled as constant PQ loads with annual 1-minute kW and kvar load profiles. Fig 2 illustrates the diversity and variability of the

active power load profiles for 20 feeder loads over a 24-hour time period.

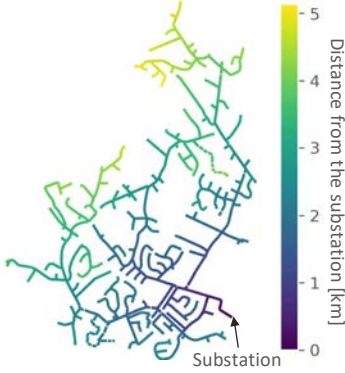


Fig 1. EPRI Ckt5

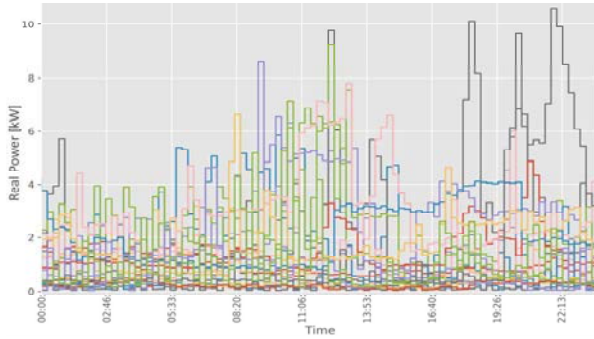


Fig 2. Real power for 20 customers (each color is a unique customer) over a 24-hr period

The load variability in Fig 2 causes voltage variability, as illustrated for a medium-voltage and a low-voltage node in Fig 3. As expected, the voltage fluctuates more for the low-voltage node as compared to the medium-voltage node.

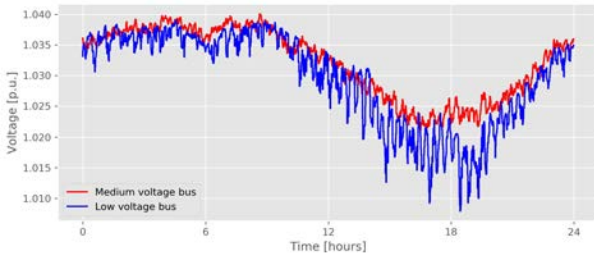


Fig 3. Voltages of a medium voltage and a low voltage node over 24 hours

### C. Modeling Loads at Lower Time Granularities

To evaluate the QSTS simulation accuracy with the lower time granularities (5-min, 15-min, and 60-min), it was necessary to emulate how AMI would record and store data at the three lower time granularities as compared to the higher 1-min time granularity. The lower time granularity load profiles were achieved by averaging the 1-min load profile values over the longer time steps of lower time granularities. For example, the average of 1-min load values between 14:15-14:30 were used to represent a load value at 15-min time granularity. Fig 4 illustrates the resulting profiles by showing an arbitrarily selected load over a 4-hour time period at the four time granularities. Note that the simulations with lower AMI granularities (5-min, 15-min, and 60-min) were conducted at their respective time step size without extrapolation.

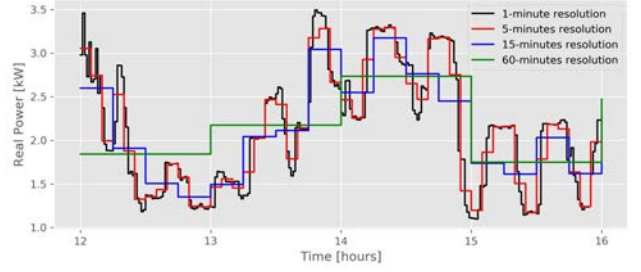


Fig 4. A load profile over a 4-hour time period at different time granularities

## III. COMPUTATIONAL CHALLENGES

The computational challenges associated with QSTS simulations are discussed in 0. In this paper, three key challenges were experienced in performing the annual QSTS simulations with high granularity of AMI real and reactive profiles for each feeder load:

**Input Data Volume:** Distinct annual 1-minute (525,600 time steps) real and reactive power profiles for each of the 1,379 loads of Ckt5 feeder formed a dataset with ~1.45 billion data points. To avoid memory issues and to effectively handle the volume of the QSTS simulation input data, the data was stored in Hierarchical Data Format (HDF) files with single-precision floating-point (32-bit/data point – 5.4Gb total vs 9.8 GB in CSV format).

**Large Number of Load Flow Solutions:** Annual QSTS simulations with 1-minute time granularity require sequentially solving 525,600 load flows. In this paper, the QSTS simulations were performed by controlling OpenDSS through the COM interface with Python scripts. Before each time step all load values were adjusted and after each time step, all nodal voltages and transformer/line phase current magnitudes were stored for later processing. All simulations were performed on a 64-bit Windows® computer with 32 Gb of RAM and four cores in an Intel® Xeon® E5-2680 processor.

**Result Data Volume:** Annual QSTS simulations with 1-minute time granularity can result in large result data volumes. In this paper, all nodal voltages and transformer/line phase current magnitudes were stored to allow comprehensive evaluation of QSTS simulation accuracy at different time granularities. The result data was stored with single-precision floating-point numbers in HDF files. At 1-minute resolution, the simulation result dataset accrued to ~13.5 GB (notably larger if stored, e.g., as comma-separated files).

## IV. SIMULATION RESULTS

The QSTS simulation results consisted of all nodal voltages and transformer/line phase current magnitudes. The results obtained with the 1-min time granularity were defined as a *reference case* and the accuracy of the QSTS simulation results with the lower (5-min, 15-min, and 60-min) time granularities was calculated as the difference to the 1-min reference case.

### A. Nodal Voltages

The nodal voltages were divided into three categories: (MV) medium voltage node, (LV1) low voltage node at the

service transformer secondary, and (LV2) low voltage node at the customer level. Ckt5 secondary circuits are modeled with a service transformer and a separate service line to each feeder load. As a result, all the low-voltage nodes in the Ckt5 model belong to either LV1 or LV2. Note that each LV2 node has a load connected to it whereas each LV1 node has around 1-5 loads connected downstream of it.

AMI data with lower time granularities does not represent fast variability in loads and thus, can miss some intermittent spikes that may result in high/low voltages and/or currents. Fig 5 illustrates the voltage magnitudes of a MV and two LV nodes around the feeder peak load with the four time granularities. As expected, lower time granularities do not capture short-duration low voltage conditions, especially on LV nodes. Also as expected, lower time granularities do not have the same impact on MV node voltages due to lower short-circuit impedances and aggregation of loads, which results in smoother load profiles. A subtle difference between LV1 and LV2 is present in this case since LV2 only has one downstream load and LV1 is at the service transformers where multiple loads (2-5) can be aggregated.

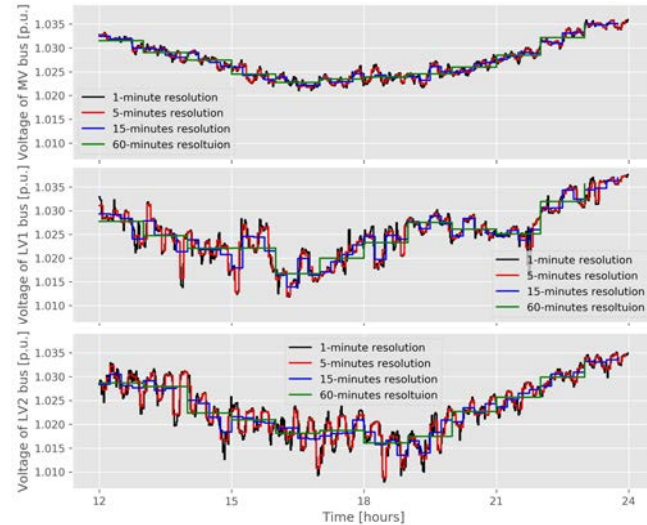


Fig 5. Voltage magnitudes for a MV and two LV nodes during peak hours with the different time granularities

Distribution engineers tend to be conservative and thus, interested in the worst-case conditions, such as lowest voltages experienced. Fig 6 shows the distribution of the lowest voltages experienced by each feeder node over the year. The grey histograms represent the reference case at 1-min time granularity and the colored histograms represent the 60-min time granularities. The arrows highlight the lowest, average, and highest values (of the distribution of minimum voltages of each node over the year). As can be seen, lower time granularity of load data does not capture short-duration load spikes and thus, results in missing short-duration voltage drops, including potential short-duration undervoltages. For instance, a small number of load nodes (LV2) experienced undervoltages ( $<0.95$  p.u.), but the lowest voltage of any node over the year with 60-min granularity was 0.9528 p.u. Similar behavior is also experienced for the service transformer secondary nodes (LV1). However, the differences between the time granularities is less profound for the MV nodes.

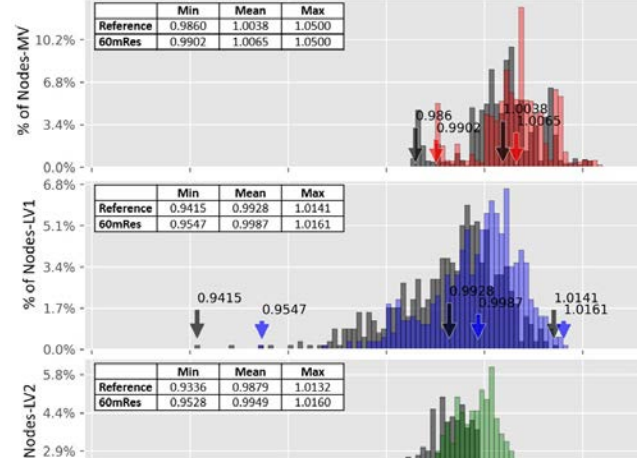


Fig 6. Distribution of the annual minimum voltages of each node for the 60-minute time granularity simulation compared to the reference. The arrows highlight the feeder minimum, average, and maximum node voltages over the year.

Extreme conditions, such as the minimum node voltages shown in Fig 6, can be infrequent and thus, may not be a major concern. Fig 7 shows distribution of the 1<sup>st</sup> percentile of the voltages of each node over the year. The distributions of 1<sup>st</sup> percentiles with 1-min and 60-min time granularities are very similar. Fig 7 suggests that the 60-min time granularity may be relatively accurate in capturing more frequent and less extreme feeder node voltages. However, as shown in Fig 6, 60-min time granularity may be insufficient to represent the extreme feeder voltage conditions.

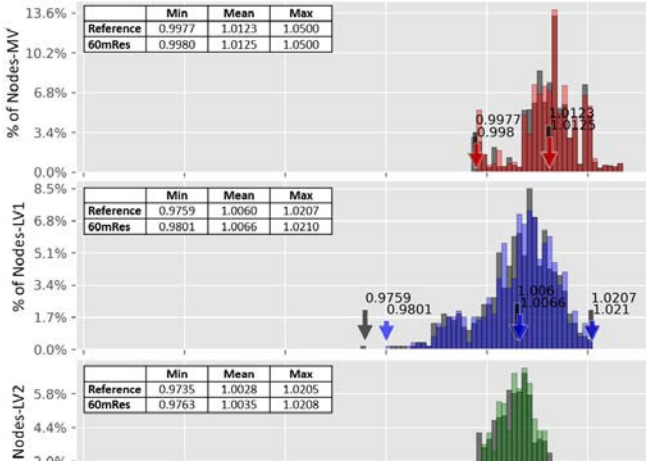


Fig 7. Distribution of the 1<sup>st</sup> percentiles of the annual voltages of each node for the 60-minute time granularity simulation compared to the reference. The arrows highlight the feeder minimum, average, and maximum node voltages over the year.

## B. Thermal Loadings

Based on the recorded phase currents, element thermal loadings were calculated. The thermal loadings were divided into three categories: MV lines, LV lines, or transformers. In the Ckt5 model, each LV line supplies only one feeder load.

Fig 8 shows the distributions of the maximum annual thermal loading of each MV line (top), LV line (middle), and service transformer (bottom) for 60-min and 1-min time granularities. As expected, lower time granularity impacts the



accuracy of the simulated maximum loadings of LV lines and service transformers more as compared to MV lines. Furthermore, some service transformers experienced notable thermal overloads at 1-min time granularity, many of which were either completely missed or significantly underestimated with the 60-min time granularity. It should be noted that some of the missed/underestimated thermal overloads may have short duration and thus, may not be of concern due to the thermal time constants of transformers. Nevertheless, it is clear that lower time granularities will not be able to accurately capture the maximum element loadings.

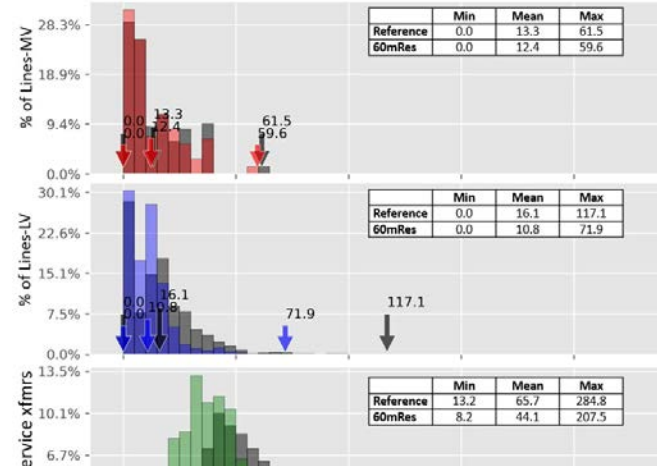


Fig 8. Distribution of the annual maximum thermal loadings of each line/transformer for the 60-minute time granularity simulation compared to the reference. The arrows highlight the feeder minimum, average, and maximum thermal loadings over the year.

Extreme conditions, such as the maximum thermal loadings shown in Fig 8, can be infrequent and thus, may not be a major concern. Fig 9 shows distribution of the 99<sup>th</sup> percentiles of the thermal loadings of each element over the year. Similar to the node voltage distributions in Fig 7, the distributions obtained with the 60-minute time granularity are similar to the 1-minute time granularity, suggesting that lower time granularities may be sufficient in capturing more frequent and less extreme feeder element thermal loadings. However, some service transformer overloads were missed, which may be important as the 99<sup>th</sup> percentile represents ~87 hours of the year.

### C. Summary Results

TABLE I lists summary results comparing the 60-minute and 1-minute time granularity results. The table lists (in parenthesis) the averages (over all nodes/elements) for the 1-minute time granularity and (in front of the parenthesis) the differences between the 60-minute and 1-minute averages.

TABLE I. The averages over all nodes/elements for the 1-minute time granularity (in parenthesis), and the differences between the 60-minute and 1-minute averages over all nodes/elements (in front of the parenthesis) for the annual minimum, 1<sup>st</sup> percentile, 99<sup>th</sup> percentile, and maximum per node or line.

| 60-Minute            | Minimum          | 1 <sup>st</sup> Percentile | 99 <sup>th</sup> Percentile | Maximum            |
|----------------------|------------------|----------------------------|-----------------------------|--------------------|
| Voltage (MV) [p.u.]  | +0.0027 (1.0038) | +0.0002 (1.0123)           | -0.0001 (1.0429)            | -0.0004 (1.0441)   |
| Voltage (LV1) [p.u.] | +0.0059 (0.9928) | +0.0005 (1.0060)           | -0.0001 (1.0434)            | -0.0005 (1.0449)   |
| Voltage (LV2) [p.u.] | +0.0070 (0.9879) | +0.0007 (1.0028)           | -0.0002 (1.0430)            | -0.0006 (1.0445)   |
| Lines (MV) [%]       | +0.1342 (1.6129) | +0.0332 (1.9496)           | -0.0890 (10.7620)           | -0.8947 (13.2517)  |
| Lines (LV) [%]       | +0.1799 (0.5441) | +0.0837 (0.8127)           | -0.9887 (8.7501)            | -5.2704 (16.0976)  |
| Transformers [%]     | +0.8259 (2.2197) | +0.4362 (3.2829)           | -4.0688 (35.6492)           | -21.5770 (65.7097) |

The average of the distribution is displayed to capture the general trend of the network. Note that the minimum and 1<sup>st</sup> percentile values for node voltages are also shown with arrows in Fig 6 and Fig 7, and the maximum thermal loadings for elements are shown with arrows in Fig 8 and Fig 9.

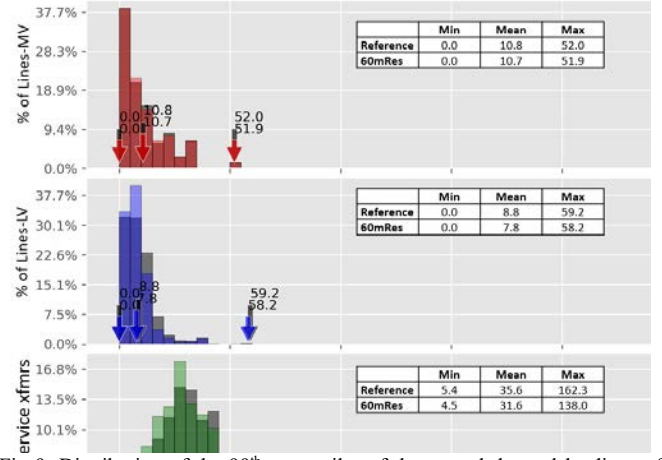


Fig 9. Distribution of the 99<sup>th</sup> percentiles of the annual thermal loadings of each line/transformer for the 60-minute time granularity simulation compared to the reference. The arrows highlight the feeder minimum, average, and maximum thermal loadings over the year.

### V. TIME GRANULARITY SENSITIVITY ANALYSIS

While Section IV focused on comparing 60-minute and 1-minute time granularities, this section expands the discussion to the two other time granularities (5-min and 15-min) that were also analyzed in this paper. Fig 10 shows the distribution outlines of the node minimum voltages (i.e., Fig 6 for all the time granularities). At 5-minute time granularity, the feeder lowest MV and LV node voltages (compare the black and red arrows) were accurately captured. At 15-minute time granularity, the feeder MV node voltages were captured accurately but the feeder LV node voltages less accurately.

Fig 11 shows the distribution outlines of the element maximum thermal loadings (i.e., Fig 8 for all the time granularities). Maximum thermal loadings of MV lines were accurately captured at all time granularities. However, maximum thermal loadings of LV lines and service transformers were less accurately captured. As expected, the higher the time granularity, the more accurately the extreme conditions were captured. Also, the lower time granularities were less accurate in capturing the service line maximum thermal loadings as compared to the service transformer maximum thermal loadings. This is because the LV lines in this feeder model only supply one feeder load whereas the service transformers could have a handful of loads decreasing in impact of individual load spikes.

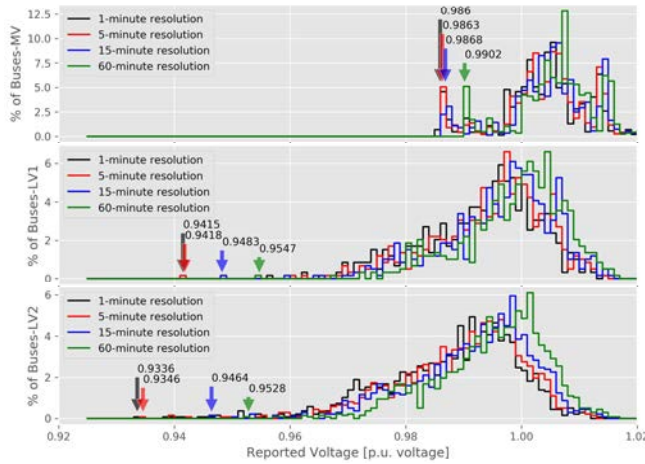


Fig 10. Outline of the distributions of the annual minimum voltages of all nodes for the four analyzed time granularities

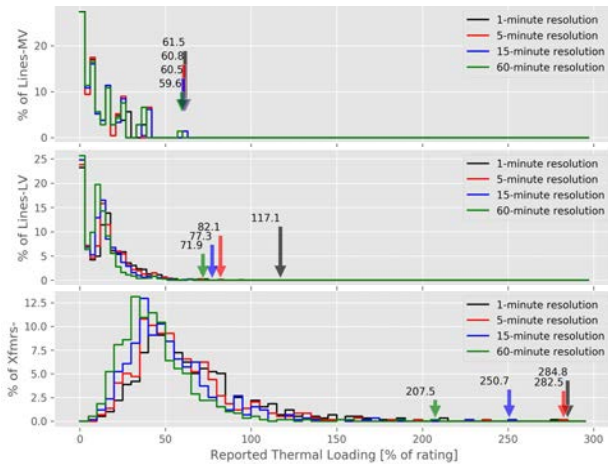


Fig 11. Outline of the distributions of the annual maximum thermal loadings of all elements for the four analyzed time granularities

Although higher time granularity in the load data provide a more accurate simulation, the benefit of additional data is negated by the computational burdensome associated with big data. Since the resolution at which the simulation is conducted is directly related to the granularity of the AMI data, the memory requirement from both the input (load profiles) and output (voltage and thermal loading measurements) of the simulation grows rapidly with higher granularity. Thus, a simulation at 15-min time granularity would require 1/15<sup>th</sup> of the memory required for the 1-min simulation, which can be a significant factor in the decision for the time granularity of the simulation, based on the purpose of the study in the first place.

## VI. CONCLUSIONS

While AMI provides a new level of visibility to distribution system loads, it comes at a cost. High volumes of data that must be effectively recorded, communicated, stored and processed for applications in distribution planning and operations. The volume of AMI is directly proportional to the time granularity at which the data is recorded and stored. While lower time granularities result in less burdensome data management, lower time granularities also provide less detailed view of the feeder loads. This paper analyzes how the time granularity of AMI load data impacts the accuracy of

QSTS simulation based on year-long datasets. Three time granularities (5-minute, 15-minute, and 60-minute) common to today's AMI deployments are compared against a higher time granularity of 1-minute. To isolate the impact of the load data time granularity, no PV systems or voltage regulating devices are considered in this study.

With 60-minute time granularity, the feeder's general operation was fairly well captured, but the feeder minimum node voltages and maximum element thermal loadings were poorly captured. With 15-minute time granularity, the feeder minimum node voltages and maximum element thermal loadings were fairly well captured at the medium-voltage level but less so at the low-voltage level. Finally, with 5-minute time granularity, the feeder minimum node voltages and maximum element thermal loadings were fairly well captured at both medium-voltage and low-voltage level. Even with 5-minute time granularity, some extreme low-voltage circuit conditions can be missed but the general trend is that 5-minute time granularity is fairly adequate for modeling low-voltage secondary circuit conditions. As few utilities currently employ AMI data for load modeling, future research should investigate the value of AMI data as compared to common utility load modeling practices.

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